Retinoblastoma RNA-seq: Differential Expression and Pathway Analysis

Re-analysis of a DESeq2 results table with pathway enrichment and SQL-based exploration

Summary

Retinoblastoma is a childhood eye cancer driven by loss of the *RB1* tumour suppressor. This report examines a DESeq2 differential expression table from public dataset GEO GSE125903, which compares retinoblastoma tumours with adult retina (57,773 genes, 19,952 with a valid adjusted p-value) to test whether the data show the biology expected from *RB1* loss. Using thresholds of adjusted $p < 0.05$ and at least a 2-fold change, **2,168 genes are up and 1,738 are down**. Upregulated genes are enriched for E2F targets, the G2M checkpoint, MYC targets, DNA replication and DNA repair. Downregulated genes are enriched for phototransduction and sensory perception. These results agree with known retinoblastoma biology and with an independent preranked GSEA. The analysis is a re-analysis that recovers known biology, not a new discovery.

1. Background

The RB1 protein normally restrains the E2F family of transcription factors, which turn on genes needed for DNA replication and cell division. When RB1 is lost, E2F activity is released and proliferation genes become highly expressed. MYCN amplification is also reported in aggressive retinoblastoma. Retinoblastoma tumours arise from developing retinal cells, so the normal retinal gene programme, including photoreceptor and phototransduction genes, may be reduced. The question here is whether the expression table shows these two signatures.

2. Data and methods

Step	Detail
Input	Public dataset GEO GSE125903 (Rajasekaran et al. 2019): retinoblastoma tumours vs adult retina, Illumina HiSeq 2500. DESeq2 results table: baseMean, log2 fold change, lfcSE, Wald statistic, p-value, adjusted p-value (57,773 Ensembl genes)
Filtering	Genes with a valid adjusted p-value (19,952). The rest have zero counts (5,609), were flagged as outliers (about 4,090), or were removed by independent filtering of low-expression genes (about 28,100).
Significance	Adjusted $p < 0.05$ and absolute log2 fold change at least 1
Annotation	Ensembl ID to HGNC symbol with the MSigDB v7.5.1 mapping (89% of tested genes)
Over-representation	One-sided hypergeometric test, up and down lists separately. Background: tested genes with a symbol (17,744). Gene sets of 10 to 500 genes. Benjamini-Hochberg FDR.
GSEA	Preranked on the Wald statistic, 1,000 permutations, gene sets of 15 to 500 genes (gseapy)
Gene sets	MSigDB Hallmark, KEGG, Reactome, GO Biological Process
Database	Results stored in SQLite and queried with SQL

3. Results

3.1 Differential expression

Of 19,952 tested genes, 3,906 pass both thresholds: 2,168 up and 1,738 down. A further 253 are significant but change by less than 2-fold. The upregulated side is stronger, with more genes and larger fold changes (strongest hits at adjusted p near 10^{-38} and $\log_2$ fold change of 6 to 9). The most significant upregulated genes are mostly mitotic: *DLL3, CKAP2L, NDC80, KIF20A, CENPH, GTSE1, NCAPG*. The most significant downregulated genes include the photoreceptor genes *NRL* and *GRK1*.

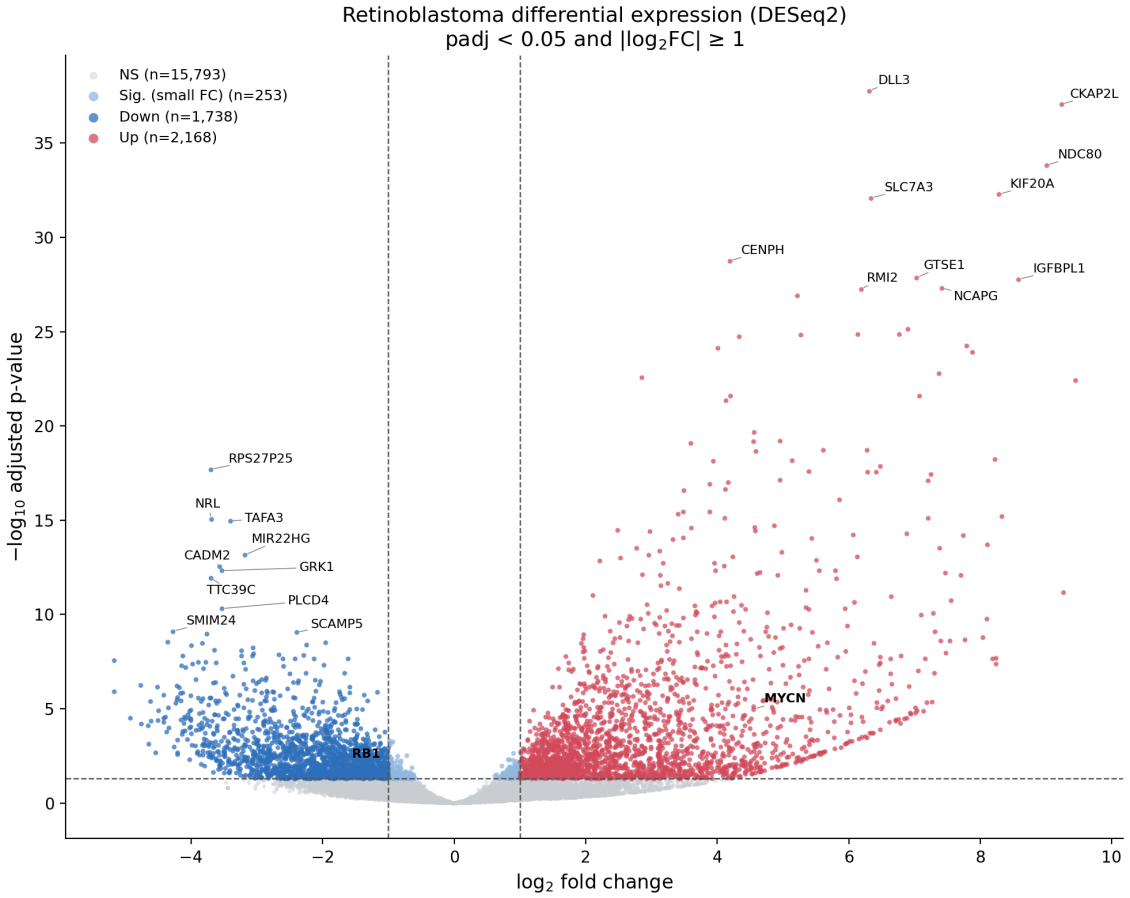


Figure 1. Volcano plot. Red: up (n = 2,168). Blue: down (n = 1,738). Dashed lines: adjusted p = 0.05 and 2-fold change.

3.2 Known retinoblastoma genes

As a sanity check, genes with known roles move in the expected direction.

Gene	Role	log2FC	Adj. p	Direction
RB1	Tumour suppressor	-1.68	8.7e-3	Down
MYCN	Oncogene	+4.55	1.1e-5	Up
MKI67	Proliferation marker	+2.77	3.1e-2	Up
TOP2A	DNA replication	+3.93	8.5e-4	Up
NRL	Rod photoreceptor	-3.69	8.7e-16	Down
GRK1	Phototransduction	-3.54	4.7e-13	Down
RHO	Phototransduction	-4.65	7.4e-4	Down

3.3 Pathway enrichment

Over-representation analysis shows two opposite programmes. Among upregulated genes, **E2F targets** (135 of 198 genes, FDR 3 x 10^{-81}) and the **G2M checkpoint** (116 of 198, FDR 4 x 10^{-59}) dominate, followed by MYC targets, mitotic spindle and DNA repair. KEGG cell cycle (FDR 1.5 x 10^{-15}) and DNA replication (FDR 2.6 x 10^{-15}) are also top terms. Among downregulated genes, Reactome **visual phototransduction** (FDR 7.7 x 10^{-11}) and GO **sensory perception of light stimulus** (FDR 1.6 x 10^{-23}) are strongest.

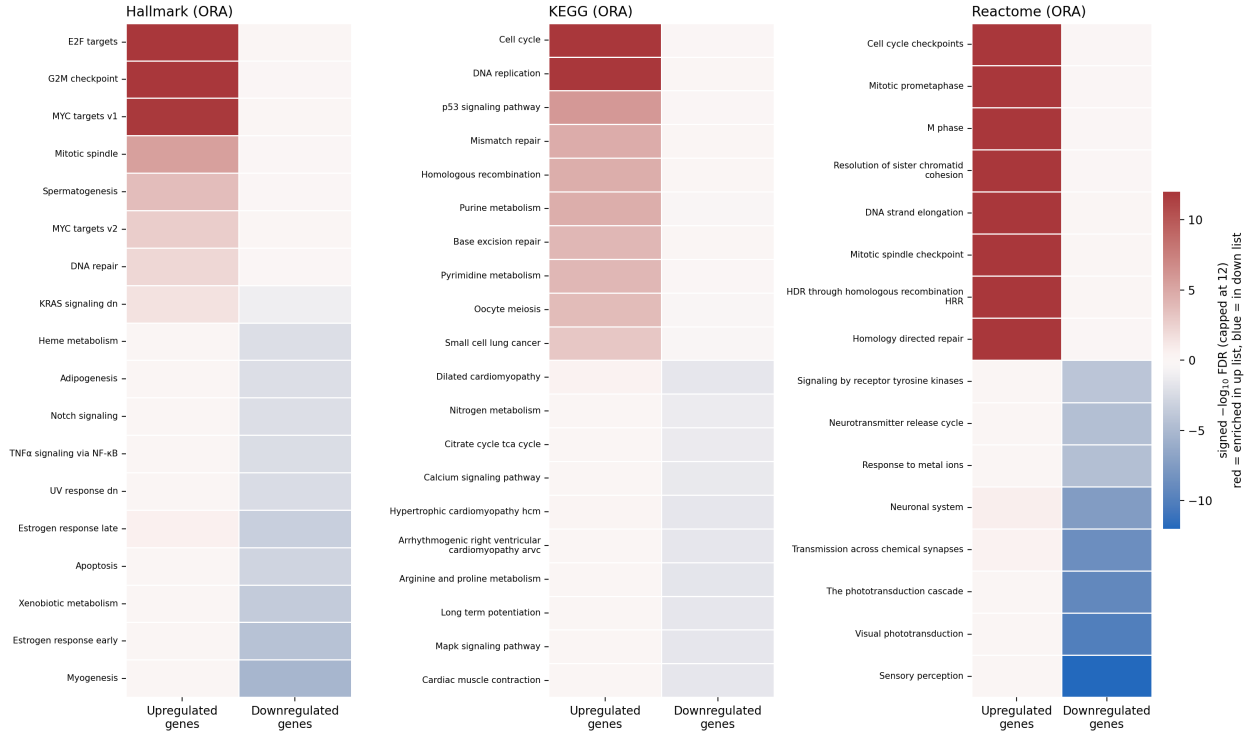


Figure 2. Enrichment of Hallmark, KEGG and Reactome terms for the up and down gene lists (signed -log10 FDR, capped at 12).

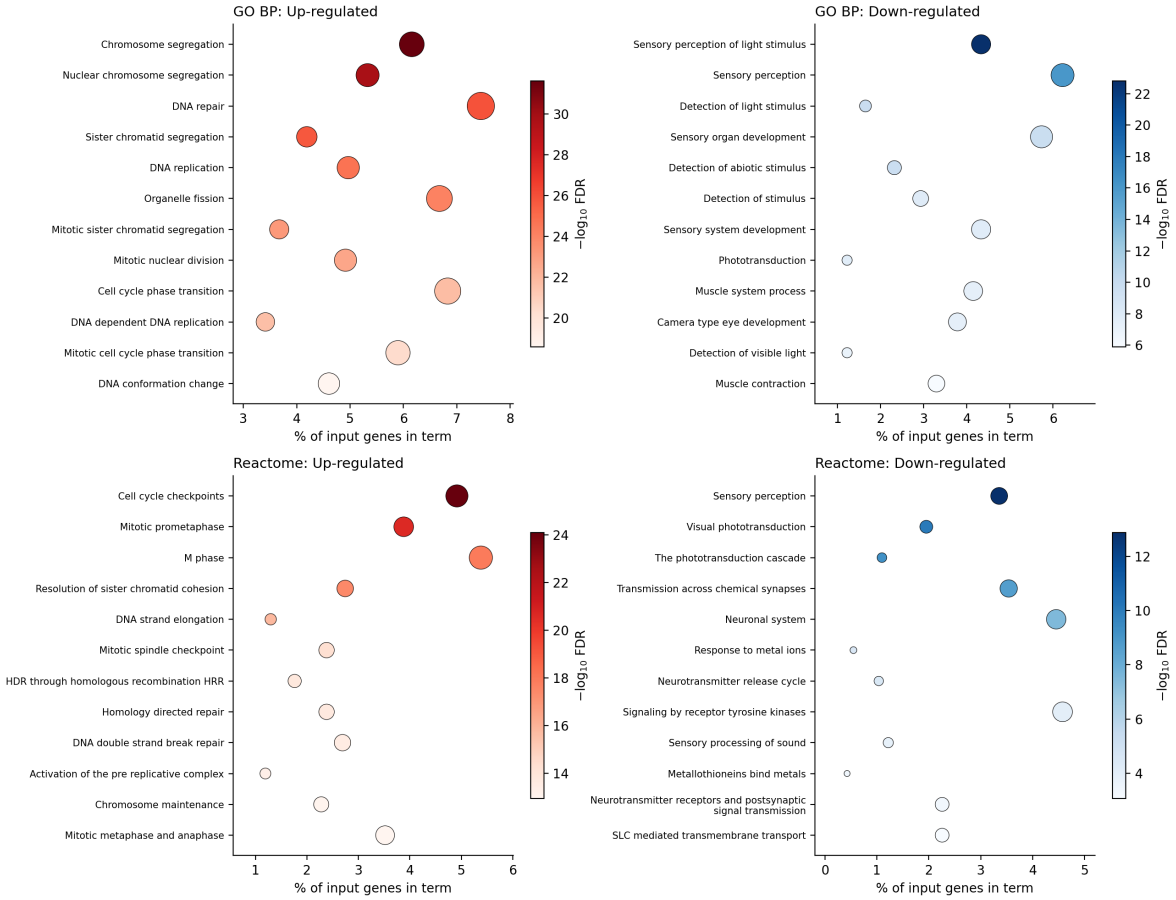


Figure 3. Top GO Biological Process and Reactome terms. Dot size: number of genes in the term. Colour: -log10 FDR.

3.4 Preranked GSEA

GSEA uses all genes ranked by the Wald statistic and does not depend on a threshold. It agrees with the over-representation result: E2F targets (NES 3.54), G2M checkpoint (3.35) and MYC targets V1 (2.75) are positively enriched, whereas the phototransduction cascade (NES -2.55) and oxidative phosphorylation (Hallmark NES -2.16) are negatively enriched. For Hallmark, 26 sets are significant at FDR below 0.05 (8 positive, 18 negative).

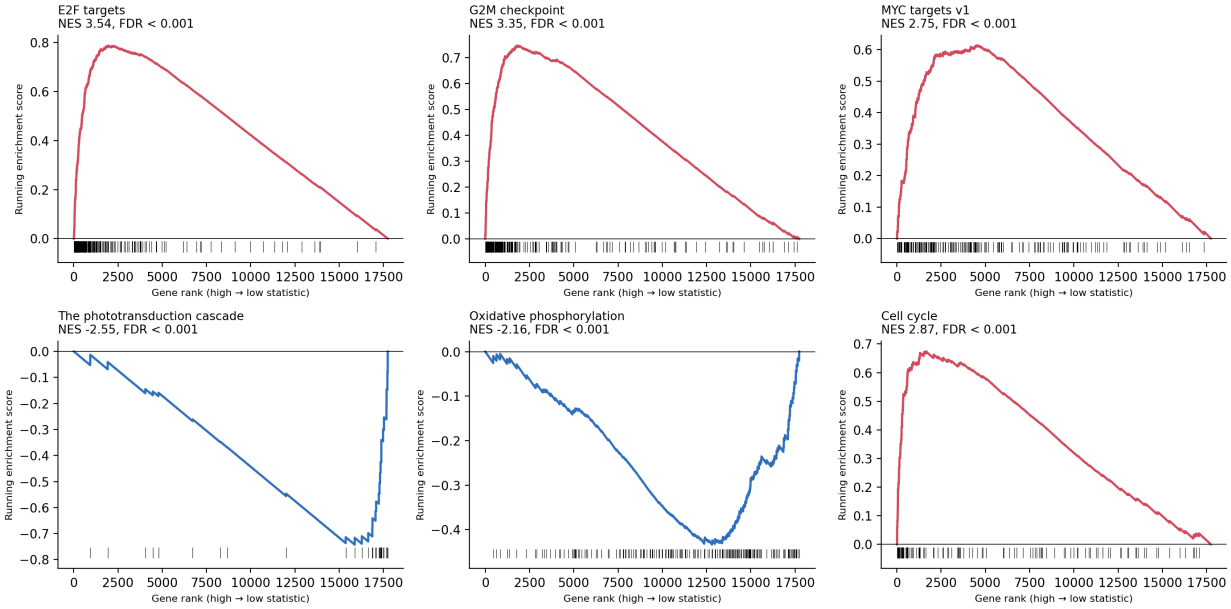


Figure 4. GSEA running enrichment scores. Red curves: positively enriched. Blue curves: negatively enriched.

3.5 Genes behind the pathways

The heatmap lists the most strongly changed genes in each key pathway. Cell-cycle genes such as *NDC80, BUB1B, E2F1, MCM2* and *MCM6* are strongly up, while the visual-cycle genes *RHO, GNAT1, PDE6B* and *GRK1* are down.

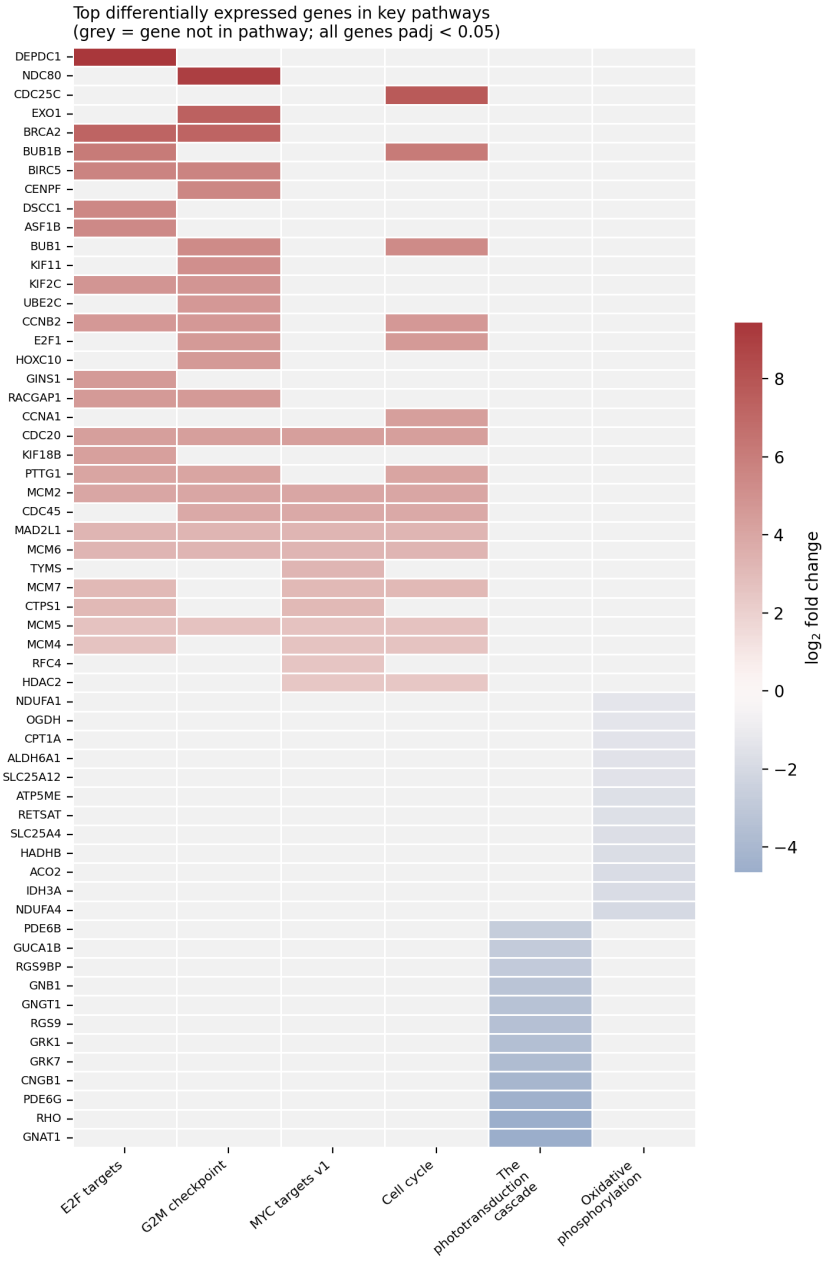


Figure 5. log2 fold change of the strongest genes in selected pathways (grey: gene not in the pathway; all genes adjusted p < 0.05).

3.6 Quality control

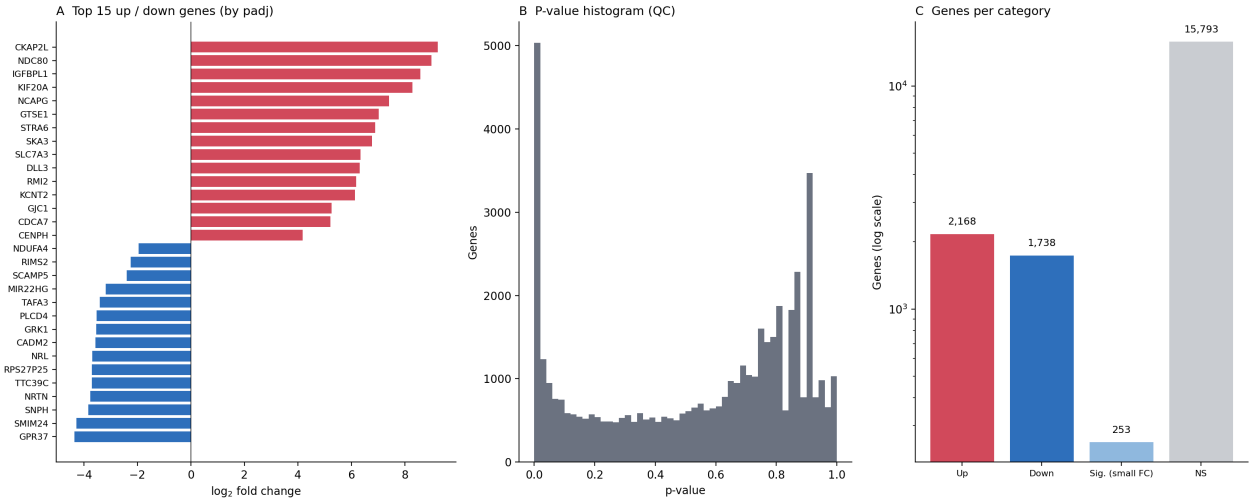


Figure 6. (A) Top 15 up and down genes by adjusted p-value. (B) P-value histogram. (C) Genes per category (log scale).

The p-value histogram shows a strong spike near zero, as expected when many genes are truly differentially expressed. The pile-up towards p = 1 comes from low-count genes with discrete counts; DESeq2 independent filtering removes most of these before adjustment.

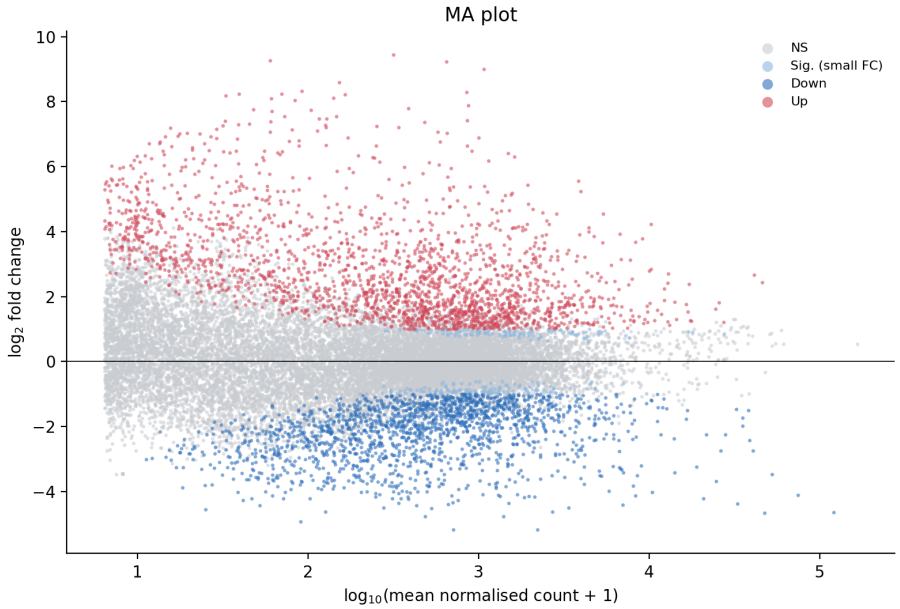


Figure 7. MA plot: mean expression against log2 fold change.

3.7 Sample-level view

The GEO raw counts file contains 7 of the 10 series samples: 5 retinoblastoma tumours and 2 adult retina (GSM3584494 and GSM3584495, labelled from the GEO sample list). Counts were converted to log2 counts per million. PCA separates tumours from retina along PC1, which carries 74.6% of the variance. The two retina samples show very high photoreceptor genes (*RHO*, *SAG*, *GNAT1*, *NRL*), very low proliferation genes (*MKI67*, *NDC80*) and higher *RB1* and lower *MYCN* than the tumours, which is the pattern expected from adult retina. The 40 strongest DE genes from the DESeq2 table split the samples in the same way. As a consistency check, the DESeq2 log2 fold changes correlate with the tumour-minus-retina difference computed from these counts (Spearman rho = 0.87, 15,341 genes), so positive log2 fold change means higher in tumour. The groups are small (5 tumours, 2 retina), so these plots illustrate the contrast and do not replace the DESeq2 test.

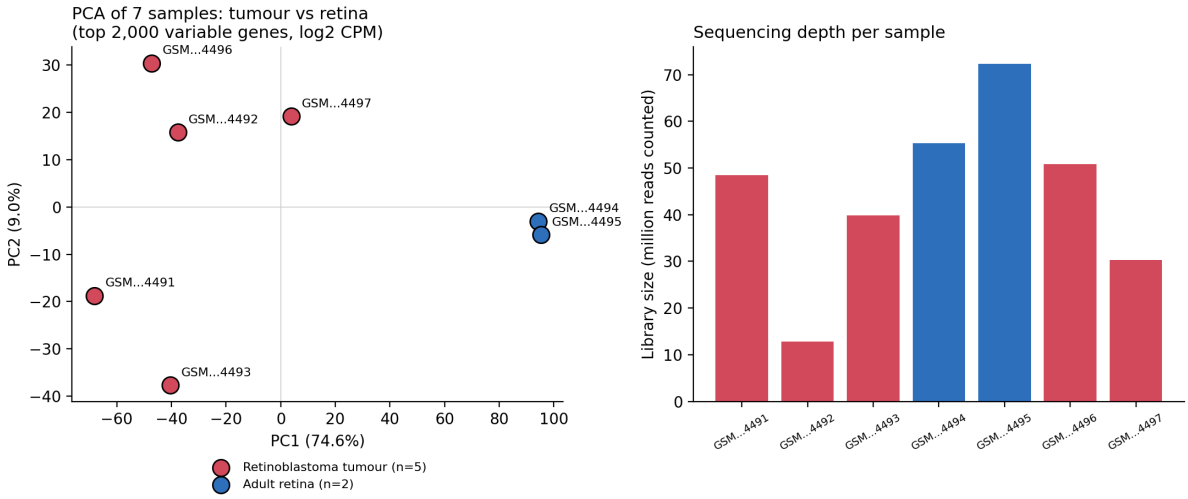


Figure 8. PCA of the 7 samples and sequencing depth per sample (red: tumour, blue: adult retina).

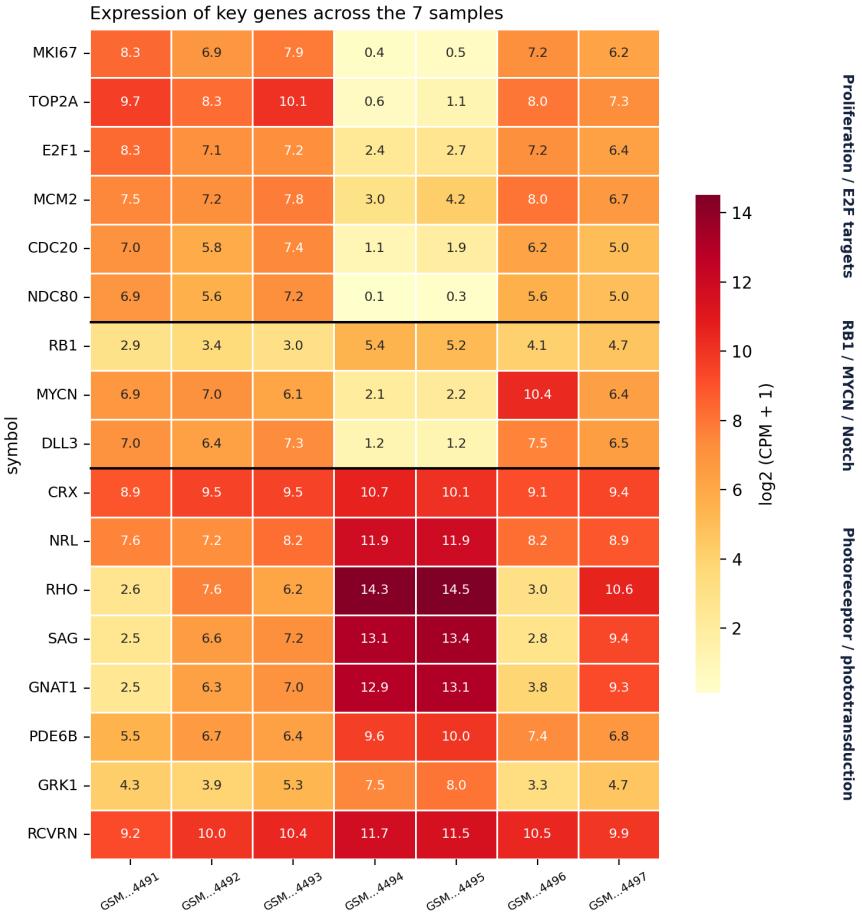


Figure 9. log2 CPM of key genes in each sample.

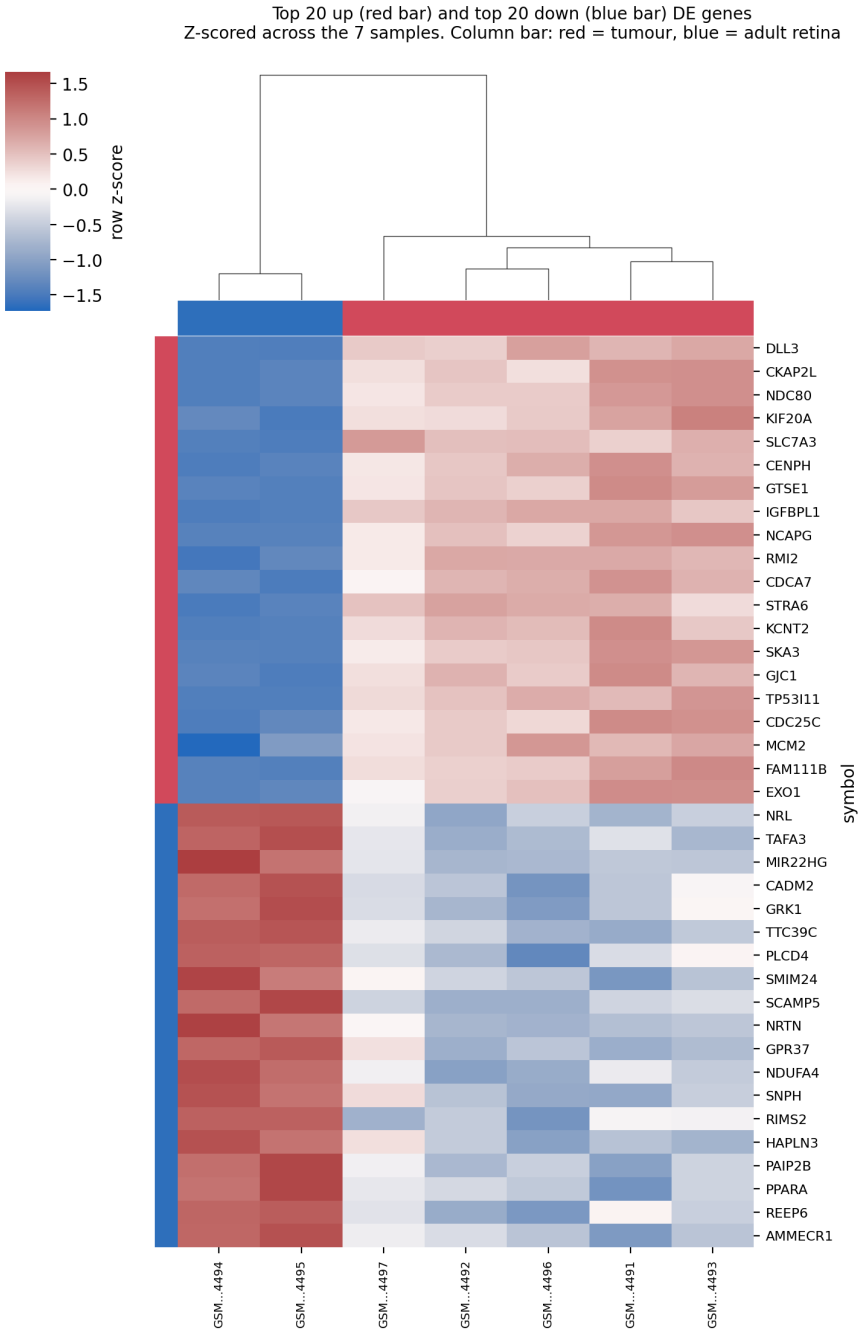


Figure 10. Top 20 up and top 20 down DE genes across the 7 samples (row z-score).

4. SQL analysis

The results were loaded into a SQLite database with four tables (`genes`, `enrichment_ora`, `enrichment_gsea`, `hallmark_membership`). Seven queries explore them with joins, CTEs, window functions, aggregation and binning. Full outputs are in `sql/query_results.md`. One example finds the Hallmark pathways with the highest mean fold change:

```
SELECT m.pathway, COUNT(*) AS genes_tested,
       SUM(g.status = 'Up') AS n_up,
       ROUND(AVG(g.log2fc), 2) AS mean_log2fc
FROM hallmark_membership m
JOIN genes g ON g.symbol = m.symbol
GROUP BY m.pathway HAVING COUNT(*) >= 30
ORDER BY mean_log2fc DESC LIMIT 5;
```

Pathway	Genes tested	Up	Mean log2FC
E2F targets	198	135	+1.99
G2M checkpoint	198	116	+1.77
MYC targets V2	56	16	+0.94
MYC targets V1	199	60	+0.80
Spermatogenesis	101	26	+0.73

5. Discussion

The data show a coherent picture. Loss of RB1 releases E2F, and E2F target genes are the single most enriched set in both methods. MYCN is strongly up and MYC target sets are enriched, which fits the role of MYC family signalling in aggressive retinoblastoma. In parallel, genes that define specialised retinal cells, especially photoreceptor phototransduction genes, are lower in the tumour. The oxidative phosphorylation signal in GSEA is also lower, which may reflect the loss of metabolically active photoreceptors. The result is a proliferative, DNA-replication-heavy profile with reduced retinal identity.

6. Limitations

- **Tissue composition (main caveat):** the comparison group is adult retina, which is rich in photoreceptors, and the two retina samples show very high photoreceptor genes. The photoreceptor and phototransduction decrease therefore largely reflects tissue composition and developmental stage, not only tumour-intrinsic repression.
- **Contrast direction** is not stored in the input table, but its log2 fold changes correlate with the tumour-minus-retina difference computed from the GEO counts (Spearman rho = 0.87), so positive values mean higher in tumour.
- **Annotation:** 11% of tested genes had no gene symbol and were excluded from enrichment.
- **Noisy fold changes:** median lfcSE is about 0.6. Applying lfcShrink (apeglm) is advisable before ranking genes by effect size.
- **Sample size and coverage:** the counts file has 7 of the 10 series samples (5 tumours, 2 retina). The groups are small, and the DESeq2 table may have used all 10 samples, so the sample plots illustrate the contrast but do not reproduce the DESeq2 test.
- **Non-specific hits:** some downregulated Hallmark sets (for example estrogen response, myogenesis) are broad signatures and should be interpreted cautiously.

- **Gene-set version:** MSigDB v7.5.1; newer releases differ slightly.

7. How this analysis was made

The Python analysis scripts and SQL queries were generated with AI assistance (Claude, Anthropic). The project owner framed the question, supplied the DESeq2 table, reviewed the outputs, checked them against known retinoblastoma biology and wrote the interpretation. The DESeq2 statistics were produced before this project. Code should be re-run and verified before reuse.

References

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- Love MI, Huber W, Anders S (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* 15:550.
- Subramanian A et al. (2005). Gene set enrichment analysis. *PNAS* 102:15545-15550.
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- Fang Z, Liu X, Peltz G (2023). GSEApy: a comprehensive package for performing gene set enrichment analysis in Python. *Bioinformatics* 39:btac757.
- Benjamini Y, Hochberg Y (1995). Controlling the false discovery rate. *J R Stat Soc B* 57:289-300.